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Genome-wide identification and expression analysis of *CBF/DREB1* gene family in *Medicago sativa* L. and functional verification of *MsCBF9* affecting flowering time

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Abstract

Background The C-repeat binding factor (CBF)/dehydration-responsive element binding (DREB1) belongs to a subfamily of the AP2/ERF (APETALA2/ethylene-responsive factor) superfamily, which can regulate many physiological and biochemical processes in plants, such as plant growth and development, hormone signal transduction and response to abiotic stress. Although the *CBF/DREB1* family has been identified in many plants, studies of the *CBF/DREB1* family in alfalfa are insufficient.

Results In this study, 25 *MsCBF* genes were identified in the genome of alfalfa ("Zhongmu No. 4"). These genes were distributed on chromosomes 1, 5, 6 and unassembled scaffolds. Phylogenetics divided the *CBF* members of *Medicago sativa*, *Arabidopsis thaliana*, and *Medicago truncatula* into six groups, of which group VI had the most *MsCBFs* members, reaching 52% (13/25). Gene duplication analysis showed that 64% (16/25) of *MsCBFs* formed tandem duplications, and 32% (8/25) formed segment duplications. The expression pattern of *MsCBF9* under different hormone treatments was verified by RT-qPCR, and it was found that *MsCBF9* responded to GA3, IAA, SA, and MeJA. Overexpression of *MsCBF9* in *Arabidopsis* significantly delayed the flowering time of *Arabidopsis*. In contrast, the flowering time of *the cbfs* mutant was earlier, and overexpression of *MsCBF9* also increased the number and size of *Arabidopsis* rosette leaves.

Conclusion In this study, the *CBF/DREB1* family of alfalfa was comprehensively identified and analyzed, and the function of *MsCBF9* in regulating flowering time was studied. This study laid a foundation for further analysis of the function of the *CBF* family in alfalfa.

Clinical trial number Not applicable.

Keywords Alfalfa, *MsCBF*;genome-wide, Flowering regulation

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Background

Transcription factors can regulate the expression of downstream genes by specifically binding to cis-acting elements in the promoter region [1, 2]. Therefore, more research has been done on the role of transcription factors in plants. AP2/ERF (APETALA2/ethylene-responsive factor) transcription factor is one of the most important plant transcription factor families [3, 4]. Among them, *DREB* (dehydration-responsive element binding) belongs to a subfamily of *AP2/ERF* superfamily, and contains only one AP2 domain [5–7]. The *DREB* subfamily can specifically recognize DRE/CRT *cis*-acting elements or core sequences with DRE elements (CCGAC) and mainly plays a role in cold stress and osmotic stress [8, 9].

Previous studies identified 56 members of the *DREB* subfamily in *Arabidopsis* and divided them into A1-A6 groups [10].*CBF/DREB1* belongs to the A1 group of the *DREB* subfamily. This group of members has an iconic sequence with a nuclear localization signal (NLS), PKKRPAGR×KF×ETRHP, upstream of the AP2 domain and a conserved DSAWR motif downstream [11]. *CBF/DREB1* has complex functions and can regulate many physiological and biochemical metabolic processes in plants, such as regulating plant growth and development, hormone signal transduction, and responding to various stresses [12–19]. They play an important role in maintaining the normal life cycles of plants.

Plants have a suitable transition time from vegetative to reproductive growth, which can optimize sexual reproduction and productivity [20]. It is well known that hormone signaling can affect plant flowering by regulating the expression of flowering-related genes. For example, salicylic acid (SA) is involved in the regulation of *CO*, *FLC*, *FT* and *SOC1* transcription, and SA-deficient *arabidopsis* shows a late flowering phenotype [21]. In rice, defects in the JA signaling pathway affect floret opening and spikelet development [22]. And GAs can regulate flower development by inhibiting the function of DELLA protein [23].Therefore, it is important to explore the regulation of flowering-related genes and their pathways. In recent years, *CBF/DREB1* has been found to regulate flowering time in plants [24–27]. Overexpression of *OsDREB1C* in rice can cause rice flowers to flower earlier and regulate the expression of flowering-related genes, such as *OsFTL2* and *OsMADS14* [24]. Overexpressing *DgDREB1A* in *Arabidopsis* delayed flowering and showed stronger freezing and drought tolerance [25]. Overexpression of *GhDREB1* in *Arabidopsis* delayed flowering in a GA-dependent manner, and flowering-related genes in different regulatory pathways were also affected by *GhDREB1* [26].

Alfalfa (*Medicago sativa* L.) is an important high-quality forage crop with high economic and ecological value [28]. It has the advantages of high adaptability, high protein content, and rich nutritional value [29, 30]. Alfalfa can be cut several times per year, but with the maturity of alfalfa, its yield increases, and forage quality decreases [31, 32]. Therefore, it is particularly important to select the appropriate cutting time. Suppose the flowering time of alfalfa can be delayed. In that case, the feed quality can be maintained for a longer period of time, and the harvest time can be extended to facilitate management without serious loss of feed quality [33–35]. *CBF/DREB1* family members have been identified in a variety of species, including 6, 12, 14 and 10 in *Arabidopsis thaliana* [10], *Lolium perenne* [36], *Glycine max* [37] and *Oryza sativa* [38], respectively. Although the *DREB* gene family in alfalfa has been preliminarily analyzed, CBF/DREB1 has not been comprehensively analyzed and the function of family members has not been explored [39]. In order to fill this gap, this study aims to identify and analyze the *CBF/DREB1* gene family of alfalfa, and comprehensively describe the characteristics of *MsCBF* genes through chromosome localization, phylogenetic analysis, conserved domain analysis and cis-acting element analysis. The function of *MsCBF9* in regulating plant flowering time was also explored. This study provides a potential reference for understanding the characteristics of the alfalfa *CBF/DREB1* family and its regulation of flowering time.

Results

Identification of *CBF* genes in the alfalfa genome

Twenty-five *MsCBFs* were identified in the alfalfa genome (*M. sativa* L. "Zhongmu No. 4") using BLASTP, and the protein sequences with two characteristic amino acid sequences, PKK/RPAGR×KF×ETRHP and DSAWR, before and after the AP2 conserved domain, with the 14th position as V (valine) and the 19th position as E (glutamic). These members were named *MsCBF1*–*MsCBF25*, based on their chromosomal locations (Fig. 1). Chromosome localization analysis showed that 25 *MsCBF* genes were unevenly distributed on Chr1, Chr5, Chr6, and unassembled scaffold chromosomes, and 72% of the *MsCBF* genes were clustered on Chr6.

The gene ID, full CDS length, protein length, molecular weight (MW), isoelectric point (pI), and subcellular location of the 25 *MsCBF* genes are shown in Table 1 and S1. The results showed that the length of *MsCBFs* varied greatly, ranging from 396 bp (*MsCBF2*) to 1233 bp (*MsCBF22*), with 76% (19/25) of the sequences being between 500 bp and 900 bp. The protein sequences of MsCBF transcription factor family members ranged in length from 131 aa (MsCBF2) to 410 aa (MsCBF22), with a mean length of 244 aa. The molecular weight (MW) of the MsCBF proteins ranged from 14.74 kDa (MsCBF2) and 46.75 kDa (MsCBF22). The isoelectric points ranged

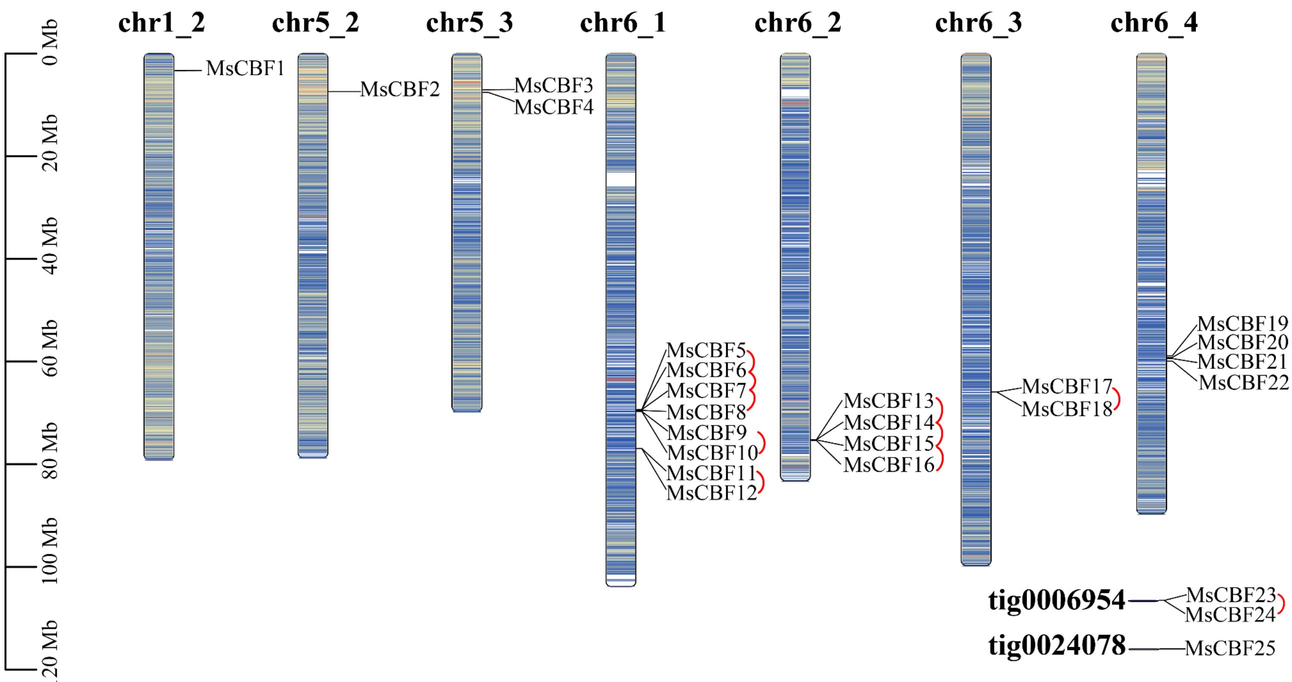

Fig. 1 Chromosome distribution of the alfalfa *CBF* gene. *MsCBFs* were mapped on chromosomes 1, 5 and 6 of alfalfa. These genes were renamed as *MsCBF1* ~ *MsCBF25* according to the order of *MsCBFs* on chromosomes. The vertical bar represents the chromosome of alfalfa, and the scale represents the chromosome length (Mb). Color gradient represents gene density, and red to blue represents high density to low density. The red semicircle represents tandem replication

from 5.1 (MsCBF12) to 9.86 (MsCBF2), with 60% (15/25) of the proteins were acidic (pI < 7.0) and 40% (10/25) were alkaline (pI > 7.0), indicating that MsCBF proteins were mostly acidic. According to the Subcellular localization prediction website, MsCBF proteins were located in the nucleus or cytoplasm, and 76% (19/25) of the proteins were distributed in both the cytoplasm and the nucleus.

Phylogenetic relationships analysis of *MsCBFs*

To understand the classification and evolutionary relationships of MsCBF in alfalfa, this study constructed phylogenetic trees of *M. sativa* and two model plants (*A. thaliana* and *M. truncatula*) using the MEGA software (Fig. 2). The results showed that 47 CBF proteins in *M. sativa* (25 proteins), *A. thaliana* (6 proteins), and *M. truncatula* (16 proteins) could be divided into six groups (Groups I–VI). The CBF members of *M. sativa* were clustered in groups I, III, V, and VI, of which 52% (13/25) were MsCBF members. Only two proteins of *A. thaliana* (AtDREB1E and AtDREB1F) can be grouped together with *M. sativa* and *M. truncatula*, and the other four proteins can be grouped together separately. In *M. truncatula*, except MtCBF7 and MtCBF14, MtCBF proteins can be grouped together with the CBF proteins of *M. sativa*. These results suggest that the CBF members of *M. sativa* and *M. truncatula* are evolutionarily closer than those of *A. thaliana*. Then, 25 MsCBFs were divided into four groups according to the phylogenetic tree, and their gene structure and motif composition were analyzed (Fig. 3A).

Gene structure and conserved motif analysis

The gene structure analysis of 25 *MsCBF* genes in alfalfa was performed by TBtools (Fig. 3B). The results showed that the number of exons in *MsCBFs* ranged from 1 to 5, the number of introns ranged from 0 to 4, and 72% (18/25) of the sequences had no more than two exons. None of the *MsCBFs* in Groups I and II had introns in their gene structure.

In order to further understand the conservation and diversity of the MsCBFs transcription factor family in evolution, NCBI-CDD search revealed that MsCBFs proteins all have only one AP2 domain (Fig. 3C). Analysis of conserved motifs (Fig. 3D) showed that members of the MsCBF transcription factor family had similar conserved motifs and arrangements. All MsCBF proteins contained Motif 1 and Motif 2, which were two conserved amino acid sequences of the AP2/ERF domain, respectively (Fig. 3E). The details of the ten motifs are provided in Table S2. Motif 1, a YRG element is the core of the AP2 domain and is involved in the specific binding of DNA. Motif 2, the RAHD element is responsible for regulating the binding strength.

Table 1 Characteristics of *MsCBF* genes in *Medicago sativa*

Gene Name	Gene ID	Full CDS length (bp)	Protein			Subcellular Location
			Length (aa)	Mw (kDa)	pI	
MsCBF1	Msa0049840	672	223	25.25	7.59	Cytoplasm. Nucleus.
MsCBF2	Msa0753830	396	131	14.74	9.86	Nucleus.
MsCBF3	Msa0795170	645	214	24.15	8.53	Cytoplasm. Nucleus.
MsCBF4	Msa0795580	645	214	24.15	8.53	Cytoplasm. Nucleus.
MsCBF5	Msa0891960	693	230	26.10	5.35	Cytoplasm. Nucleus.
MsCBF6	Msa0891970	681	226	25.74	5.59	Cytoplasm. Nucleus.
MsCBF7	Msa0891980	687	228	25.85	5.55	Cytoplasm. Nucleus.
MsCBF8	Msa0891990	777	258	29.41	7.67	Cytoplasm. Nucleus.
MsCBF9	Msa0892020	591	196	22.42	5.28	Cytoplasm. Nucleus.
MsCBF10	Msa0892030	843	280	32.11	9.2	Cytoplasm. Nucleus.
MsCBF11	Msa0893350	657	218	24.84	5.29	Cytoplasm. Nucleus.
MsCBF12	Msa0893360	603	200	22.92	5.1	Cytoplasm. Nucleus.
MsCBF13	Msa0928860	816	271	30.77	7.13	Cytoplasm.
MsCBF14	Msa0928870	912	303	34.88	7.97	Cytoplasm.
MsCBF15	Msa0928880	804	267	30.48	6.67	Nucleus.
MsCBF16	Msa0928940	492	163	18.69	6.76	Cytoplasm. Nucleus.
MsCBF17	Msa0956890	792	263	29.80	7.12	Cytoplasm. Nucleus.
MsCBF18	Msa0956900	648	215	24.56	6.66	Cytoplasm. Nucleus.
MsCBF19	Msa0992060	909	302	33.83	5.95	Cytoplasm. Nucleus.
MsCBF20	Msa0992110	678	225	25.63	6.54	Cytoplasm. Nucleus.
MsCBF21	Msa0992130	909	302	33.83	5.95	Cytoplasm. Nucleus.
MsCBF22	Msa0992230	1233	410	46.75	6.18	Cytoplasm. Nucleus.
MsCBF23	Msa1344910	876	291	33.31	6.33	Cytoplasm.
MsCBF24	Msa1344920	816	271	30.73	7.1	Cytoplasm.
MsCBF25	Msa1413960	618	205	23.27	5.41	Cytoplasm. Nucleus.

CDS: coding sequence; bp: base pair; aa: amino acid; MW: molecular weight; pI: isoelectric point

Analysis of the gene duplication and synteny of the *MsCBFs*

To better understand potential gene duplication events, TBtools was used to visualize tandem and segmental duplication events of *MsCBF* genes. The results showed that 64% (16/25) of the *MsCBFs* formed five tandem duplication events, including two tandem duplication events containing four genes (*MsCBF5*, *MsCBF6*, *MsCBF7* and *MsCBF8*) (*MsCBF13*, *MsCBF14*, *MsCBF15*, and *MsCBF16*) and four tandem duplication events containing two genes (*MsCBF9* and *MsCBF10*, *MsCBF11* and *MsCBF12*, *MsCBF17* and *MsCBF18*, *MsCBF23*, and *MsCBF24*) (Fig. 1). The collinearity analysis of the alfalfa genome showed that 32% (8/25) of the *MsCBF* genes formed four segmental duplication events on Chr6 (Fig. 4). The four pairs of segment duplication events are listed in Table S3. These results indicate that the reasons for the expansion of the *MsCBF* gene family may include tandem and segmental duplication events.

Next, collinearity maps of alfalfa with *A. thaliana* and *M. truncatula* were constructed (Fig. 5). Two *MsCBF* genes (*MsCBF2* and *MsCBF5*) were found to be involved in the formation of four homologous gene pairs in the collinearity analysis of alfalfa and *A. thaliana*, of which three pairs contained the *MsCBF2* gene. In the collinearity analysis of alfalfa and *M. truncatula*, nine *MsCBF* genes were found to be involved in the formation of eleven pairs of homologous gene pairs. The number of collinear genes between alfalfa and *M. truncatula* was approximately three times higher than that in *A. thaliana*. Gene pairs are shown in Tables S4 and S5. The above results show that the *CBF* genes of these three plants differentiated during the long-term evolution of the species, but compared with *A. thaliana*, the number of homologous pairs between alfalfa and *M. truncatula* was higher, and the *MsCBF* genes involved were wider. This may be due to the closer genetic relationship between alfalfa and *M. truncatula*, and the higher homology between *MsCBF* and *MtCBF* genes.

Prediction of *cis*-acting elements in *MsCBFs* promoters

To further clarify the biological function of the *MsCBF* genes in alfalfa, *cis*-acting elements were predicted in the promoter region (2 kb upstream of the coding region) of *MsCBF* family members using the PlantCARE database (Fig. 6). According to the prediction results, four types of elements were identified: light, growth and development, hormone, defense and stress. In detail, 84% of *MsCBF* genes contain AAGAA-motif elements, and 64% contain as-1 elements, which are related to growth and development. In defense and stress elements, all *MsCBF*

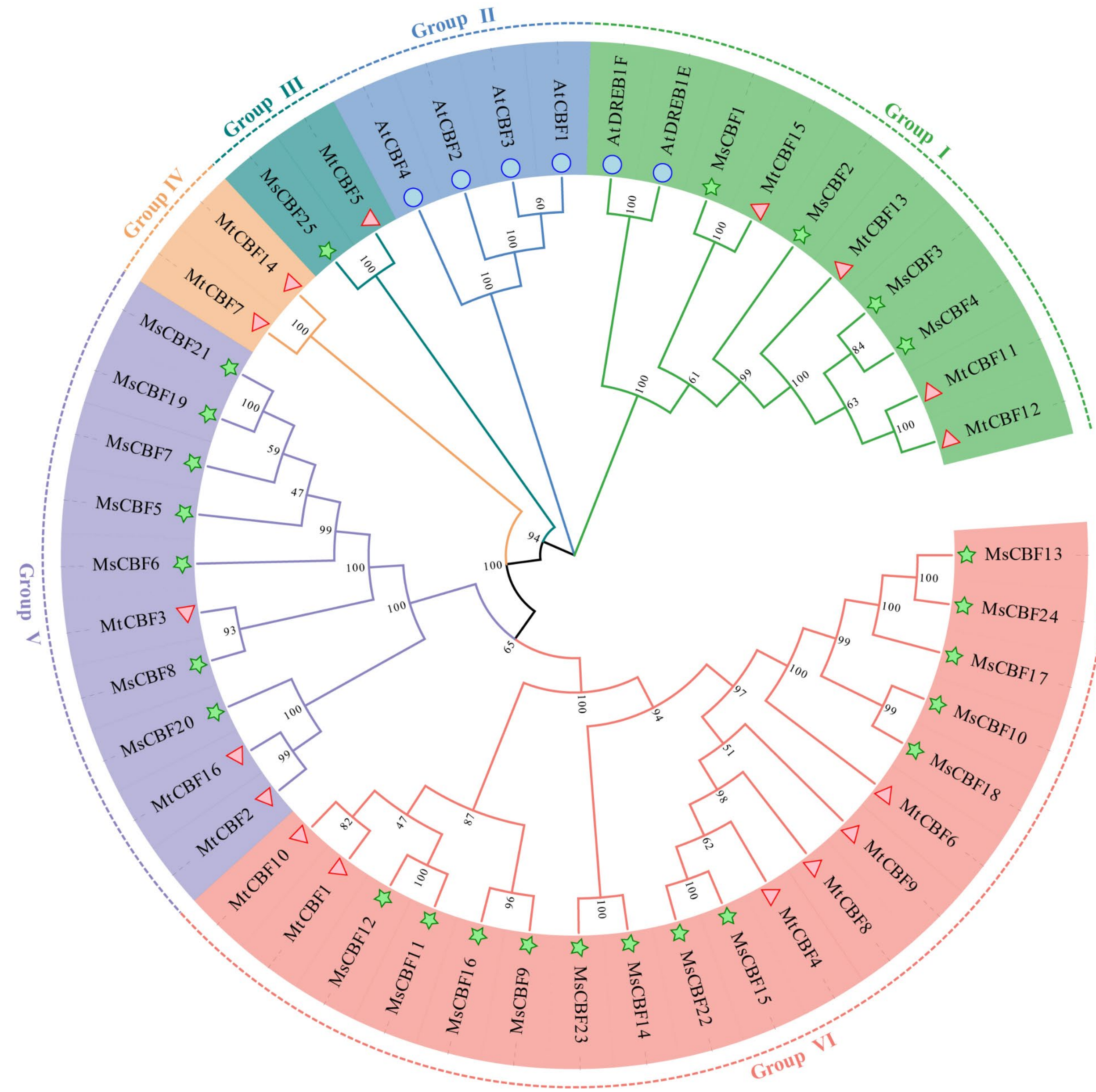

Fig. 2 Phylogenetic tree of CBF in *M. sativa*, *A. thaliana*, and *M. truncatula*. The phylogenetic tree was constructed by neighbor-joining of MEGA 6.0. The phylogenetic tree was divided into six ancestors: groups I-VI. Different markers were used to distinguish each species, with blue circles, red triangles, and green stars representing *A. thaliana*, *M. truncatula*, and *M. sativa*, respectively

members contained MYB, MYC elements, 96% members contained ARE elements, and 76% members contained STRE and WRE3 elements. Additionally, different *MsCBFs* contain different hormone response elements, indicating that the expression of these *MsCBF* genes may be induced by different hormones. These results suggest that *MsCBF* genes may respond to a variety of stress conditions and may also be involved in photoperiod- and hormone-induced growth, development, and flowering regulation.

Expression analysis of the *MsCBF9* gene under different hormone treatments

Plant hormones play important roles in the regulation of plant growth and development. Therefore, we used different hormone treatments to explore the expression patterns of *MsCBF9* (Fig. 7). After treatment with 200 μM GA3, the expression of *MsCBF9* was significantly decreased, and the expression level at 72 h was decreased by 53 times compared with the control. After treatment with 200 μM MeJA, the expression of *MsCBF9* showed a downward trend as a whole, decreasing by 160 times at

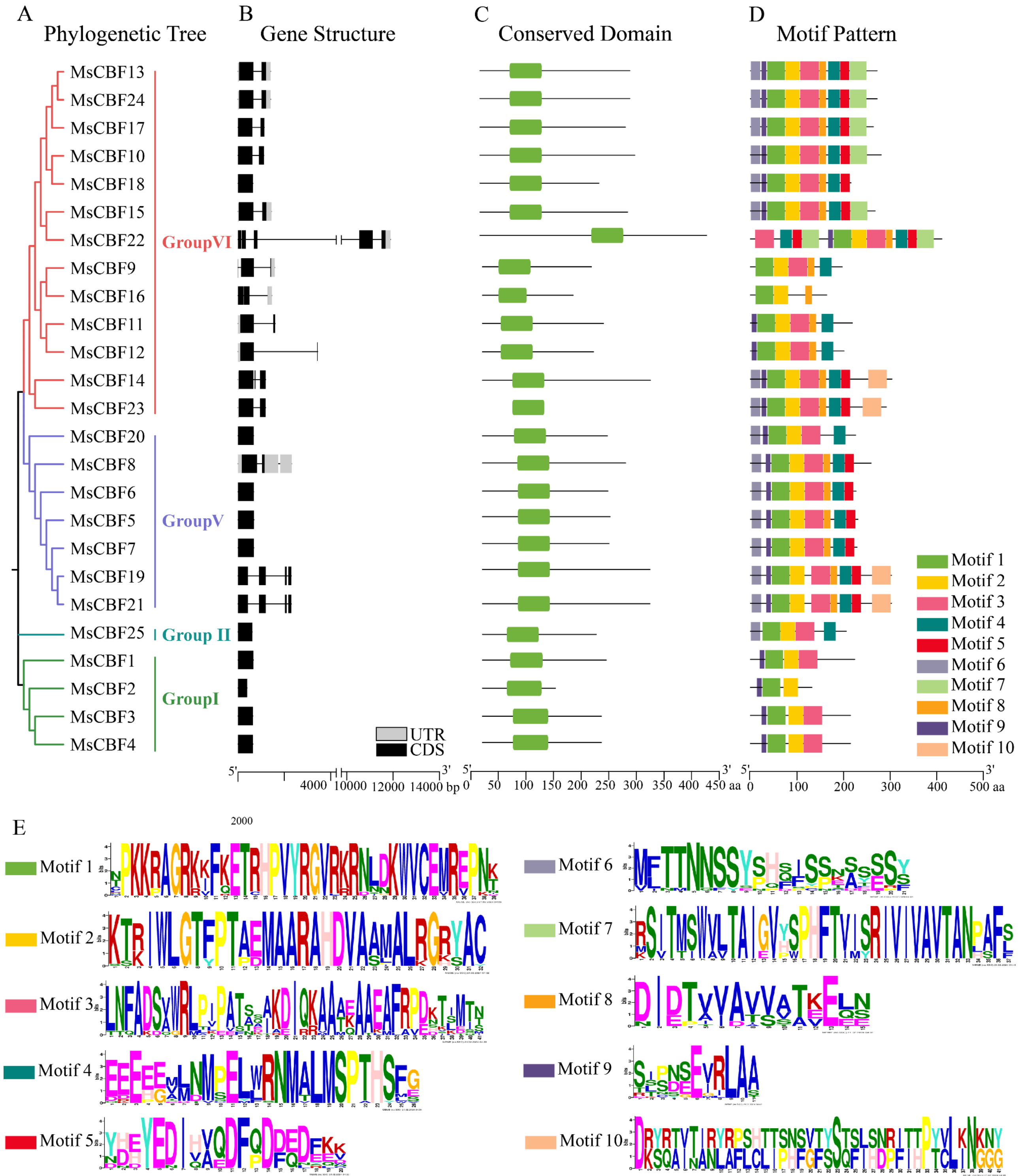

Fig. 3 Phylogenetic relationship, motif composition, and gene structure analysis of the *MsCBF* family. (**A**) The phylogenetic tree of alfalfa *MsCBFs* was constructed using the neighbor-joining method in MEGA 6.0. Different colors represent different groups. (**B**) The gene structure of *MsCBFs*. The gray box, black box, and black line represent the UTR, CDS, and intron, respectively. (**C**) AP2 conserved domain of MsCBFs. (**D**) The MEME program was used to analyze the motif composition of MsCBFs. Different colors were used for each domain. (**E**) Sequence logos of Motif 1-Motif 10

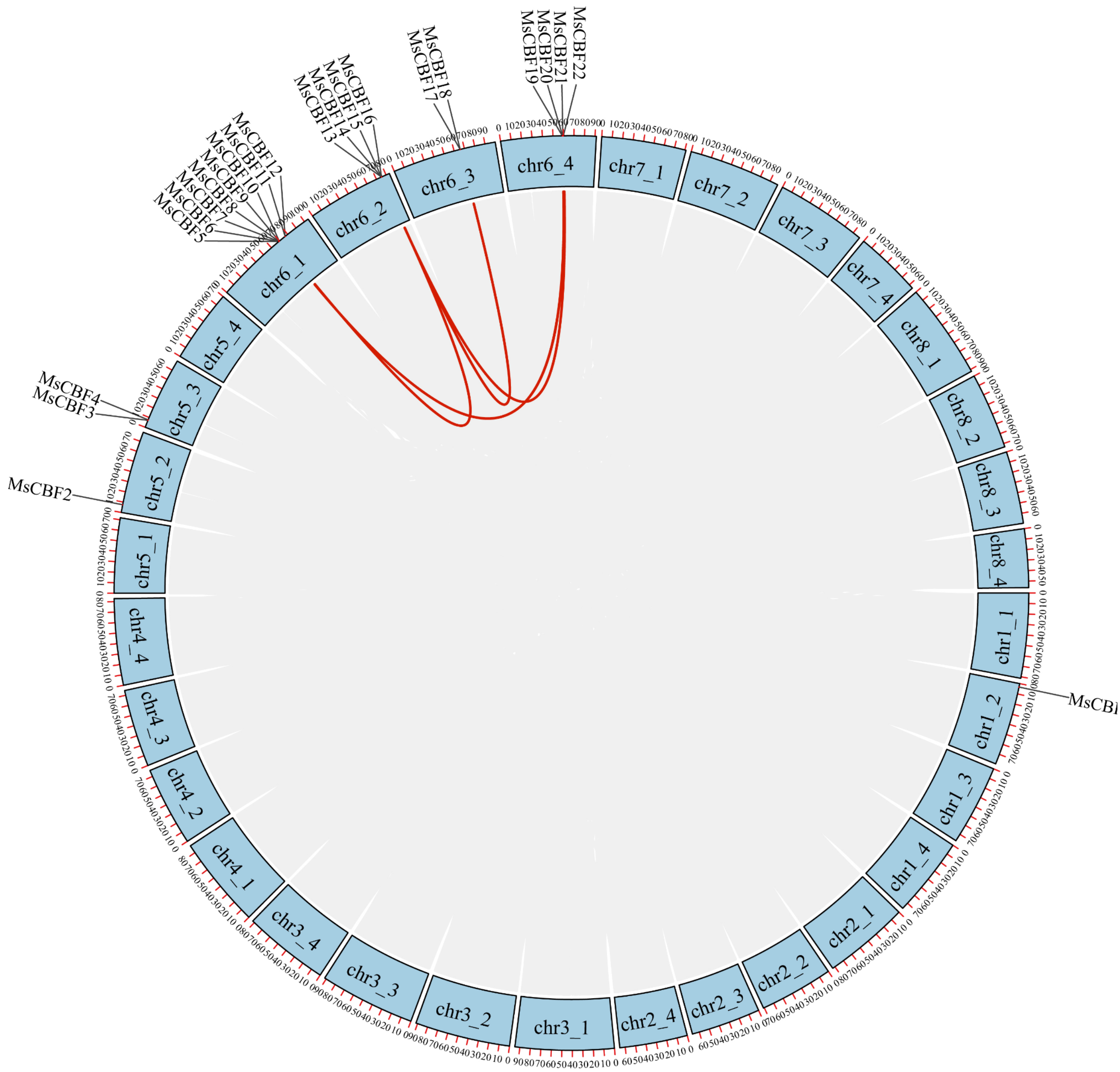

Fig. 4 The syntenic relationship diagram of *MsCBF* genes in alfalfa. The blue circle represents the alfalfa chromosome, and the position of the *MsCBF* gene is shown on the circle. The gray band represents the syntenic regions in the alfalfa genome, and the red band represents the segment replication event

72 h. After treatment with 200 μM IAA, the expression of *MsCBF9* increased first and then decreased. The expression of *MsCBF9* was significantly activated from 5 to 24 h and decreased after 48 h. After treatment with 200 μM SA, the expression of *MsCBF9* decreased significantly at 3 h, began to recover at 5 h, and decreased again at 24 h.

Flowering phenotype identification in transgenic *Arabidopsis*

In order to study the function of the *MsCBF9* gene in regulating flowering, *MsCBF9* was heterologously expressed in *A. thaliana* (Fig. 8). It was found that under long-day conditions, the flowering time of transgenic lines OE-1 and OE-20 was 1–3 days later than that of WT, and the flowering time of the *cbfs* mutant was two days earlier than that of WT (Fig. 8A, D). On the 30th day of growth, the plant height of the two overexpression lines was significantly lower than that of WT, the rosette leaves of OE-1 and OE-20 were larger, and the number of leaves was also five and three more than that of WT, respectively (Fig. 8B, C, E, F). The plant height of the *cbfs* mutant was significantly higher than that of the WT, but there was no significant difference in the number of leaf leaves between the *cbfs* mutant and WT plants. Through

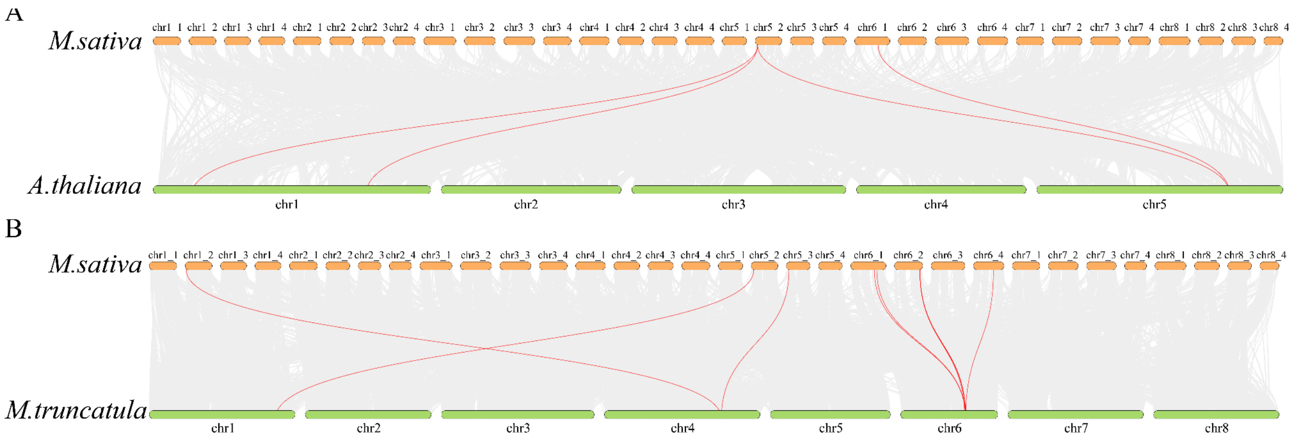

Fig. 5 Collinearity analysis of the *MsCBF* genes in *A. thaliana* and *M. truncatula*. The orange and green boxes represent the chromosomes of alfalfa and model plants, respectively. The gray line represents the syntenic block between alfalfa and the model plants, and the red line represents segmental duplication events of *CBF* gene pairs

qRT-PCR detection of flower-promoting related genes (*FT*, *LFY*, *SOC1*, *FCA*) of *A. thaliana* in transgenic lines, WT and *cbfs* mutant, it was found that these genes were significantly inhibited in transgenic lines, while significantly up-regulated in *cbfs* mutants (except *LFY* was not significantly different from WT) (Fig. 9).

Discussion

CBF/DREB1 transcription factors exist only in plants and play an important role in regulating plant growth and development and responding to abiotic stresses [8, 18, 40]. *CBF/DREB1* family members have been identified in many species, including 6, 12, 14, 6, and 10 in *A. thaliana* [10], *Lolium perenne* [36], *Glycine max* [37], *Camellia sinensis* [41] and *Taraxacum kok-saghyz* [42]. In our study, 25 *MsCBF* genes were identified in the genome of alfalfa "Zhongmu No.4", and these genes were comprehensively analyzed for protein characteristics, phylogenetic relationships, gene duplication, gene structure, conserved domains and *cis*-acting elements. The regulation of flowering time by *MsCBF9* was explored.

In this study, a phylogenetic tree of CBF proteins in *M. sativa*, *M. truncatula*, and *A. thaliana* was constructed, which could be divided into six groups: Group I-Group VI. The alfalfa *CBF* members were clustered in groups I, III, V, and VI. *MsCBF1-MsCBF4* was clustered with *AtDREB1E* and *AtDREB1F*. Previous studies have shown that *AtDREB1E* and *AtDREB1F* in *Arabidopsis* are related to salt and drought stress [43], so it is speculated that *MsCBF1-MsCBF4* may also be related to salt and drought stress. Chromosomal localization analysis showed that 11 *MsCBFs* were clustered on chromosome 6 of alfalfa. Previous studies also found that *MtCBFs* were clustered on chromosome 6 of *M. truncatula*, indicating that these genes may synergistically regulate downstream genes or functional redundancy [44]. Genome-wide duplications play an important role in the amplification of members of many gene families. Tandem duplication and segmental duplication are the two main forms of duplication [45]. In this study, we found that 64% of *MsCBFs* formed tandem duplication events, whereas only 32% formed segmental duplication events, indicating that tandem duplication may be the main driving force for expanding the *CBF/DREB1* gene family. The phenomenon in which *CBFs* genes are arranged in tandem on chromosomes has also been reported previously. For example, *CBF* genes in Triticeae genomes are arranged in tandem on chromosome 5, and *CBF* genes in *Zea mays* genomes are arranged in tandem on chromosome 7 [46, 47]. Even a relatively small number of *Oryza sativa CBF* genes show a tandem arrangement on chromosome 9 [48]. Analysis of the conserved domains of MsCBFs revealed that Motif1 and Motif2 contain the characteristic conserved elements YRG and RAHD of the AP2 domain, respectively. These two elements play important roles in the binding of CBF transcription factors to specific downstream DNA sequences [49].

Plant hormones play an important role in regulating plant growth and development, and previous studies have shown that they contribute to the regulation of plant flowering. The GA pathway is one of six plant flowering pathways [50, 51]. The exogenous addition of GA can accelerate the flowering of *Arabidopsis* under short-day conditions, and it was found that the *gai-1* mutant flowered extremely late under short-day conditions [52, 53]. Exogenous IAA can induce flower stalk elongation in *Brassia campestris* [54]. MeJA treatment can promote the opening of sorghum florets by regulating genes related to metabolic pathways and plant hormone signal transduction pathways in sorghum floret plasma cells [55]. Studies have also shown that SA is associated with many flower developmental processes, such as pollen germination and

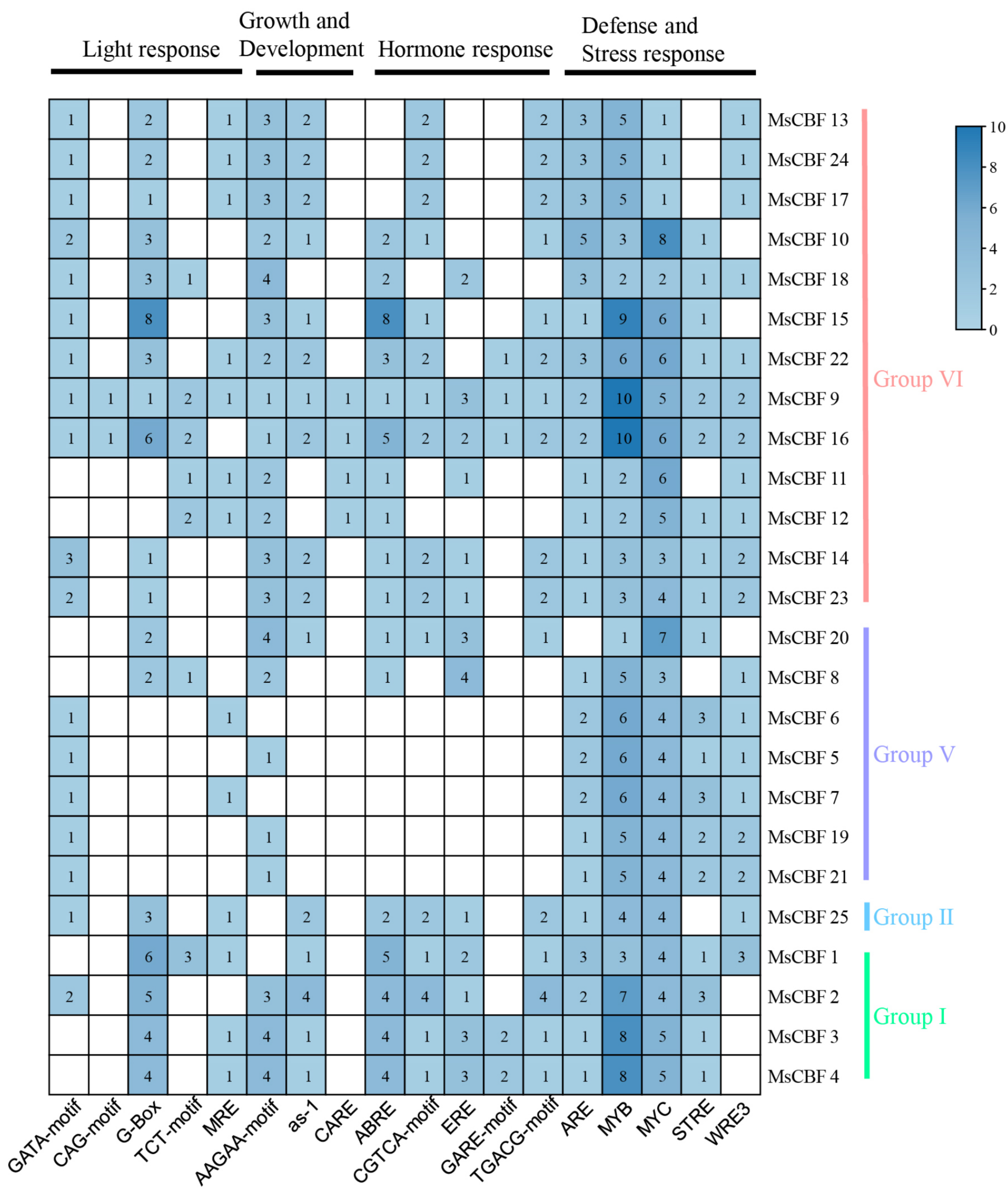

Fig. 6 Prediction of cis-acting elements of *CBF* genes promoter in alfalfa. Cis-acting elements are divided into light, growth and development, hormones, defense, and stress-related elements. The numbers in the boxes represent the number of elements. The color gradient on the right-hand side represents the number of elements. The darker the color of the box, the greater the number of elements

pollen tube elongation [56, 57]. This study also explored whether *MsCBFs* responded to various plant hormones. The results showed that *MsCBF9* responded to IAA, GA, SA and MeJA.Under the treatment of 200 μM IAA, GA3, SA and MeJA, the expression of *MsCBF9* was significantly inhibited. These results suggest that *MsCBF9* may regulate plant flowering by influencing the hormone signaling process. Next, this study heterologously expressed

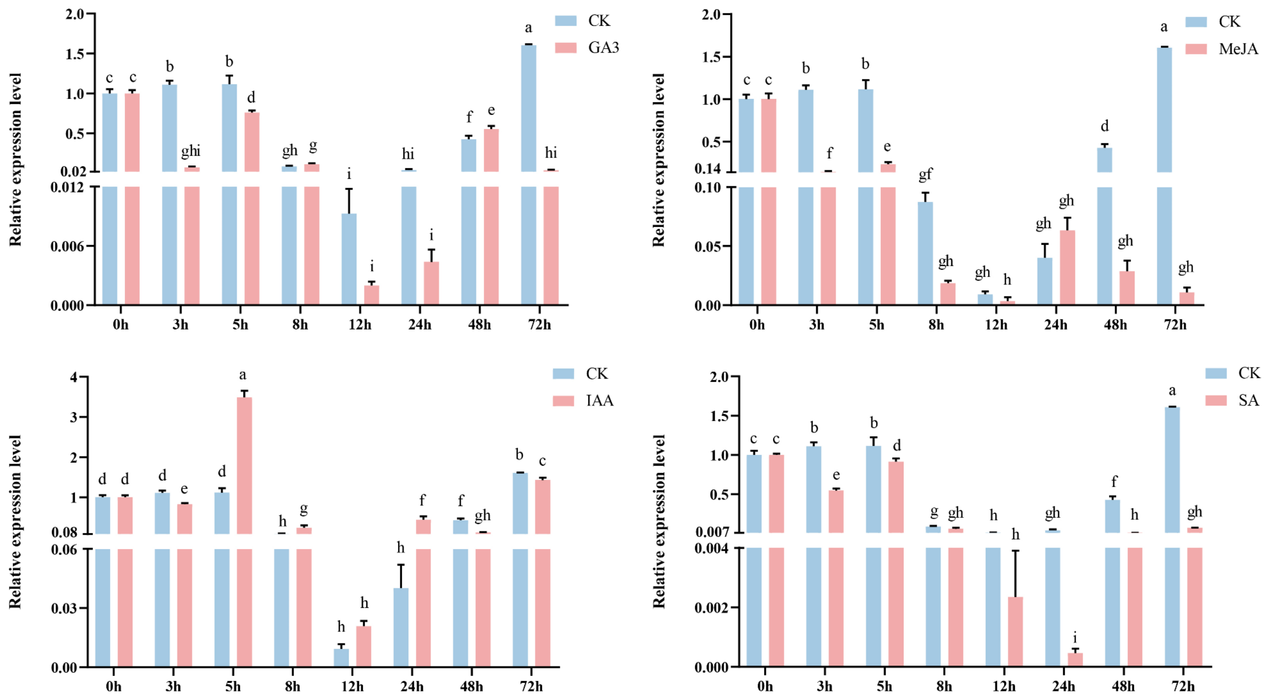

Fig. 7 The expression profiles of *MsCBF9* under 200μM GA3, MeJA, IAA and SA treatments. The horizontal coordinates represent the processing time (0 h, 3 h, 5 h, 8 h, 12 h, 24 h, 48 h, 72 h), and the vertical coordinates represent relative expression levels. The error bar represents the standard error of the mean of three independent repetitions. ANOVA with $p < 0.05$. There were significant differences between different letters (**a**-**i**)

MsCBF9 in *Arabidopsis* to further explore the regulation of this gene during flowering. Overexpression of *MsCBF9* in *Arabidopsis* significantly delayed the flowering time of *Arabidopsis*, while the flowering time of *cbfs* mutants was significantly advanced. Overexpression of *MsCBF9* also increased the number of rosette leaves in *Arabidopsis*. Previous studies have shown that *CBF* genes can delay flowering in plants. For example, heterologous expression of *LcCBF2* and *LcCBF3* in *Arabidopsis* can lead to delayed flowering [58]. It has been reported that the *CBF* gene causes flowering delay, which may be partially attributed to its activation of *FLC* and subsequent inhibition of the two flowering pathway integrators *FT* and *SOC1* [59]. This finding is consistent with the results of the present study. The expression of *FT* and *SOC1* genes in the two overexpression lines was significantly inhibited, while the expression of these two genes in the *cbfs* mutant was significantly increased. Other flowering-related genes, such as *LFY* and *FCA*, have the same expression pattern.

Conclusions

A total of 25 *MsCBF* genes were identified in this study, which were mainly distributed on chromosomes 1, 5 and 6. Among them, 64% of the genes formed tandem repeats, indicating that tandem duplication may be the main driving force for *MsCBFs* amplification. Subcellular localization prediction showed that MsCBF protein was located in the nucleus and cytoplasm. Further analysis of the domains shows that *MsCBF* members located in the same branch of the evolutionary tree contain similar domains. Moreover, each MsCBF contains YRG and RAHD elements to ensure that the CBF protein can function normally. Through the analysis of *MsCBFs* promoter, it was found that *MsCBFs* contained many cis-acting elements related to hormones, growth and development. *MsCBF9* was selected for subsequent research, and it was found that the gene responded to four hormones: IAA, GA, SA and MeJA. Heterologous expression of *MsCBF9* in *Arabidopsis* can significantly delay the flowering time of *Arabidopsis*, while the flowering time of *cbfs* mutant plants is significantly earlier. This study provides preliminary insights into the molecular basis of the *CBF* gene family in alfalfa. However, the mechanism by which *MsCBF9* regulates flowering and the functions of other *MsCBF* members are still unclear. Therefore, future research will focus on the analysis of the function and specific regulatory mechanisms of *MsCBFs.*

Methods and materials

Identification of *CBF* Family members in alfalfa

The CBF/DREB1 protein sequences of *A. thaliana* were obtained from TAIR website (https://www.arabidopsis.org/, accessed on 8 January 2024). The genome and annotation information of alfalfa (Cultivar "Zhongmu

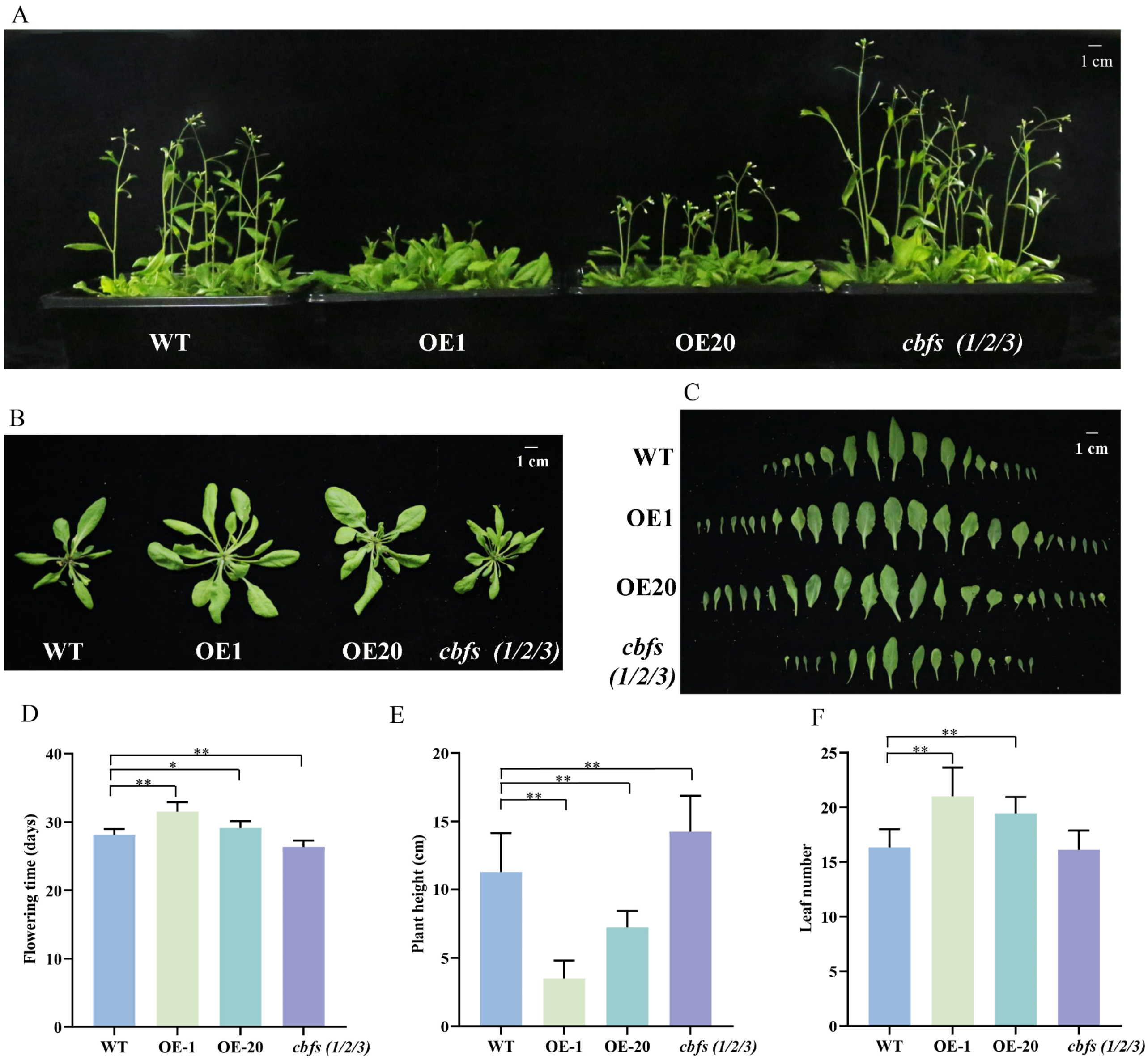

Fig. 8 Regulation of flowering period by overexpression of *MsCBF9* in *Arabidopsis* under long-day conditions. (**A**) The status of WT, overexpressing *MsCBF9* lines OE-1, OE-20 and *cbfs* mutants at 30 days of growth. (**B**) (**C**) The size and morphology of *Arabidopsis* rosette leaves of WT, overexpressing *MsCBF9* lines OE-1, OE-20, and *cbfs* mutants at 30 days of growth. (**D**) WT, overexpressing *MsCBF9* lines OE-1, OE-20 and *cbfs* mutant flowering time statistics. Each line had 9 biological replicates. (**E**) Plant height statistics of WT, overexpressing *MsCBF9* lines OE-1, OE-20, and *cbfs* mutants at 30 d of growth. Each line had 9 biological replicates. (**F**) The number of rosette leaves of WT, overexpressing *MsCBF9* lines OE-1, OE-20, and *cbfs* mutants was counted after 30 days of growth. Each line had 9 biological replicates. The asterisk in the column indicates a significant difference between the WT and transgenic lines (*$p < 0.05$; **$p < 0.01$, t-test)

No.4”) was downloaded from the figshare data repository (https://figshare.com/s/fb4ba8e0b871007a9e6c, accessed on 8 January 2024). The CBF/DREB1 protein sequences of *M. truncatula* were obtained from EnsemblPlants website (https://plants.ensembl.org/index.html, accessed on 8 January 2024). AtCBF1 (AT4G25490), AtCBF2 (AT4G25470), AtCBF3 (AT4G25480), AtCBF4 (AT5G51990), AtDREB1E (AT1G63030), AtDREB1F (AT1G12610) protein sequences were used as query sequences to perform BLASTP in TBtools software with 1×10^{-10} cutoff E-values against the alfalfa reference genome. Then, NCBI-CDD (https://www.ncbi.nlm.nih.gov/cdd, accessed on 21 January 2024) was used to search for protein sequences containing only one AP2 domain. Clustal Omega (https://www.ebi.ac.uk/jdispatcher/msa/clustalo, accessed on 21 January 2024) of EBI (European Bioinformatics Institute) was used for multiple sequence alignment to find two characteristic amino

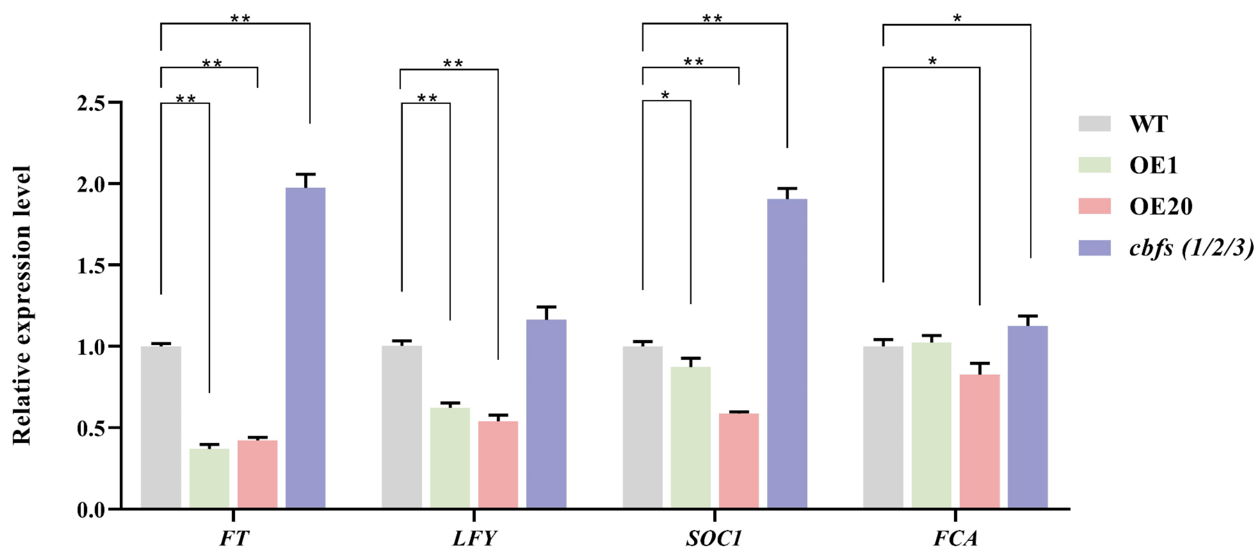

Fig. 9 Expression levels of flowering-related genes (*FT*, *LFY*, *SOC1*, and *FCA*) in *Arabidopsis thaliana*. The horizontal coordinates represent different genes, and the vertical coordinates represent the relative expression levels. The error bar represents the standard error of the mean of three independent repetitions. The asterisk in the column indicates a significant difference between the WT and transgenic lines (*$p < 0.05$; **$p < 0.01$, t-test)

acid sequences PKK/RPAGR×KF×ETRHP and DSAWR before and after the AP2 conserved domain, and the 14th V (valine) and 19th E (glutamic acid) in the AP2 domain. The protein sequences with these characteristics belong to the *CBF/DREB1* gene family. Expasy (https://www.expasy.org/about) was used to remove redundancy of the obtained MsCBF protein sequence, with default values for all parameters. Finally, 25 *CBF* family members were identified in alfalfa. The predicted isoelectric (pI) and molecular weights (MWs) of all *MsCBF* genes were predicted with the Expasy-ProtParam website (https://web.expasy.org/protparam/, accessed on 29 January 2024). Plant-mPLo (http://www.csbio.sjtu.edu.cn/bioinf/plant-multi/, accessed on 29 January 2024) was used to predict subcellular localization.

Chromosome location analysis and phylogenetic tree construction

The genomic location of the identified *MsCBF* was confirmed and mapped using Gene Location Visualize from the GTG/GFF of TBtools software (v1.108, Chen, C., GZ, China) [60]. Using the ClustalW program in MEGA 6.0 software (Tamura, K., Tokyo, Japan) [61], full-length amino acid sequences were aligned, and a phylogenetic tree was constructed with CBF protein sequences from alfalfa, *M. truncatula*, and *A. thaliana* using the neighbor-joining (NJ) method (bootstrap values for 1000 replicates, other parameters are selected by default). The iTOL website was used to beautify phylogenetic trees (https://itol.embl.de/, accessed on 3 February 2024).

Gene structure and conserved domains analysis

The MEME online program (https://meme-suite.org/meme/doc/meme.html, accessed on 8 February 2024) was used to analyze conserved motifs in MsCBF proteins, and the maximum number of motifs was 10 [62]. The structural (intron-exon) information of *MsCBFs* were included in the alfalfa GFF file. Both conserved motifs and gene structures were visualized using TBtools (v1.108, Chen, C., GZ, China).

Cis-acting elements analysis of *MsCBFs* promoter

The 2 kb upstream of the coding region of *MsCBF* genes was obtained from the alfalfa genome database. Extracted sequences were submitted to the PlantCARE website (http://bioinformatics.psb.ugent.be/webtools/plantcare/html/, accessed on 24 February 2024) for *cis*-acting element analysis, and visualized using TBtools (v1.108, Chen, C., GZ, China) [63].

Gene duplication and synteny analysis

One-step MCScanX-Super fast of TBtools (v1.108, Chen, C., GZ, China) with default parameters was used to analyze the pattern of gene duplication of the *MsCBFs.* The Dual Synteny Plot was used to determine the syntenic relationship between orthologous *MsCBF*s and other species.

Plant materials and growth conditions

The alfalfa seeds "Zhongmu No. 4" were obtained from the Institute of Animal Science of the Chinese Academy of Agricultural Sciences. The growth conditions were consistent with those previously reported [29]. After three weeks, alfalfa was treated with hormones. The experimental groups were treated with 1/2 Hoagland solution supplemented with 200 μM GA3, SA, IAA, and MeJA. The control group only added 1/2 Hoagland solution. The leaves of the seedlings were collected at 0, 3, 5, 8, 12, 24, 48, and 72 h and stored at − 80 °C.

Cloning and expression vector construction of *MsCBF9*

Total RNA was extracted from alfalfa leaves using a Plant Total RNA Extraction Kit (Promega, Madison, WI, USA), and cDNA was obtained by reverse transcription. According to the genome database of alfalfa "Zhongmu No.4", the full-length CDS sequence of *MsCBF9* (Gene ID: Msa0892020) was extracted for gene synthesis, and the *pUC57-MsCBF9* plasmid was obtained. The specific primer Super-MsCBF9-F/R, containing the homologous arm, was designed by selecting the *SmaI* restriction site, and the target sequence was amplified from the *pUC57-MsCBF9* plasmid. Primer sequences are shown in Table S7. The amplified product was recovered and seamlessly cloned into an expression vector to construct a *Super-MsCBF9-GFP* expression vector to obtain a recombinant plasmid. The correctly sequenced plasmid was transformed into Agrobacterium GV3101 (Huayueyang Biotechnology, Beijing, China) and heterologously transformed into *Arabidopsis.* Transgenic *Arabidopsis* was screened on 1/2MS medium containing 4 mg/ml phosphinothricin (PPT) (Coolaber, Beijing, China), and PCR detection was performed. Positive plants were selected for screening until the homozygous plants were obtained.

Flowering phenotype identification in transgenic *Arabidopsis*

WT, transgenic lines OE-1, OE-20, and *cbfs* mutant plants were planted on 1/2MS medium, placed at 4 °C for three days of vernalization, and moved to the incubator after three days. After ten days, it was moved into the soil. Nine seedlings were planted per pot. The first day of growth was recorded when seeds germinated on 1/2MS medium. At 30 days, the flowering status of the different lines was observed and photographed, and the plant height and number of rosette leaves were counted.

Real-time quantitative PCR (RT-qPCR) analysis

The RT-qPCR method was consistent with a previous study [64]. Primer sequences are shown in Table S6. Three technical replicates were performed for each biological repetition. RT-qPCR was used to analyze the relative gene expression using the $2^{-\Delta\Delta Ct}$ method.

Statistical analysis

SPSS software was used for statistical analysis. Duncan multiple range test was used to analyze differences between groups (ANOVA with $p<0.05$). A t-test was used to determine whether there was a significant difference between the two samples ($^{*}p<0.05$; $^{**}p<0.01$). GraphPad Prism 9 (GraphPad Software, Boston, FL, USA) and Adobe Illustrator CC 2020 (ADOBE, San Jose, CA, USA) were used for the drawing and graphic editing, respectively.

Abbreviations

CBF	C-repeat binding facto
DREB	Dehydration-responsive element binding
DREB1	Dehydration-responsive element binding 1
AP2	APETALA2
ERF	Ethylene-responsive factor
CDS	Coding sequence
bp	Base pair
aa	Amino acid
MW	Molecular weight
pl	Isoelectric point
Mb	Chromosome lengt
RT	qPCR Real-Time Quantitative PCR

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12870-025-06081-0.

Supplementary Material 1

Supplementary Material 2

Supplementary Material 3

Supplementary Material 4

Supplementary Material 5

Supplementary Material 6

Supplementary Material 7

Author contributions

Experimental design and planning and first draft writing, J.C., Y.L., J.K.; preparation and modification of the images, H.L.; data processing, manuscript modification, X.J., L.Z.; data analysis and test data accuracy, H.D., X.W.; application and analysis of the software used in the experiment, F.H., M.L.; data and manuscript review, J.K.; funding acquisition, J.K. All the authors contributed to the article. All authors have read and agreed to the published version of the manuscript.

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Data availability

All data generated or analyzed during this study are included in this published article and its supplementary information files.

Declarations

Ethics approval and consent to participate

Field and laboratory studies were conducted through local legislation. This article does not include any research involving human participants or animals, nor any endangered or protected species. The plant materials collected and the experiments conducted in this study are consistent with institutional, national and international guidelines and legislation.

Consent for publication

Not applicable.

Competing interests
The authors declare no competing interests.

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